LAB WEEK 2

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1.4 Exercises

For each of the following problems, identify the problem inputs, processes, and problem outputs. Then write pseudocode and create flowchart for each solution.

1. A program will change a temperature reading from Celsius to Fahrenheit. Prompt the user to enter the temperature in degree Celsius and display the temperature in Fahrenheit. (Fahrenheit = 32 + (9*Celsius/5))

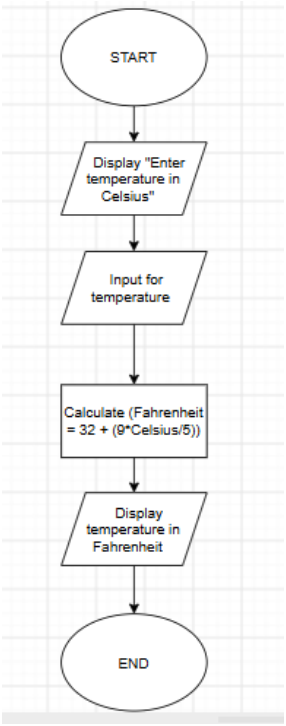
Step 1 : Analysis

a) Input : Temperature in degree Celsius
b) Process : Calculate Fahrenheit = 32 + (9*Celsius/5)
c) Output : Display the temperature in Fahrenheit

Step 2 : Pseudocode

1. Start
2. Display “Enter temperature in Celsius
3. Call input Celsius
4. Calculate fahrenheit = 32 + (9 * celsius / 5)
5. Display temperature in fahrenheit
6. End

Step 3 : Flowchart



2. Let a runner insert how many steps he has run in one minute. Find how many miles he has run. Assume each of his step length is 2.5 feet. (1 mile = 5280 feet)

Step 1 : Analysis

a) Input : Number of steps
b) Process :
   i) Calculate total feet (total feet = Steps * 2.5)
   ii) Calculate miles (miles = total_feet / 5280)
c) Output : Total distance run in miles

Step 2 : Pseudocode

1. Start

2. Display “Enter Steps”

3. Input number of steps

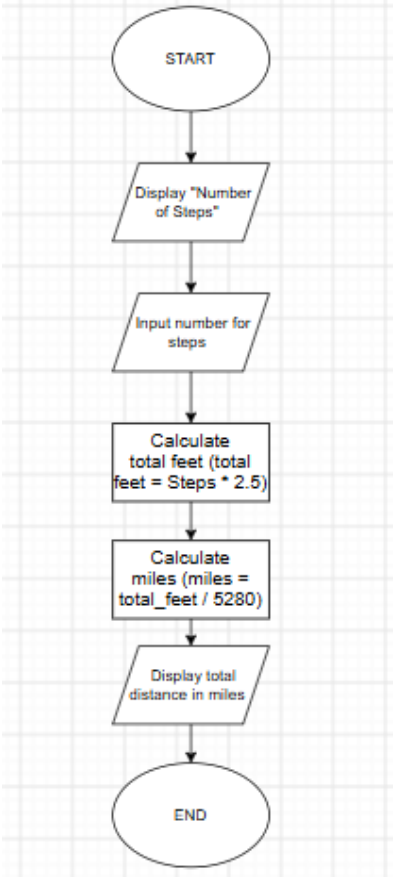
4. Calculate total feet (total feet = Steps * 2.5)

5. Calculate miles (miles = total_feet / 5280)

6. Display total distance run in miles

7. End

Step 3 : Flowchart



3. Accept two numbers and find their sum and difference. If the sum is more than the difference, display the sum otherwise display the difference.

Step 1 : Analysis

a) Input : Input num1, num2

b) Process :

i) sum = num1 + num2

ii) difference = num1 - num2

iii) Decision (Diamond): Is sum > difference?

- If Yes (True), print sum

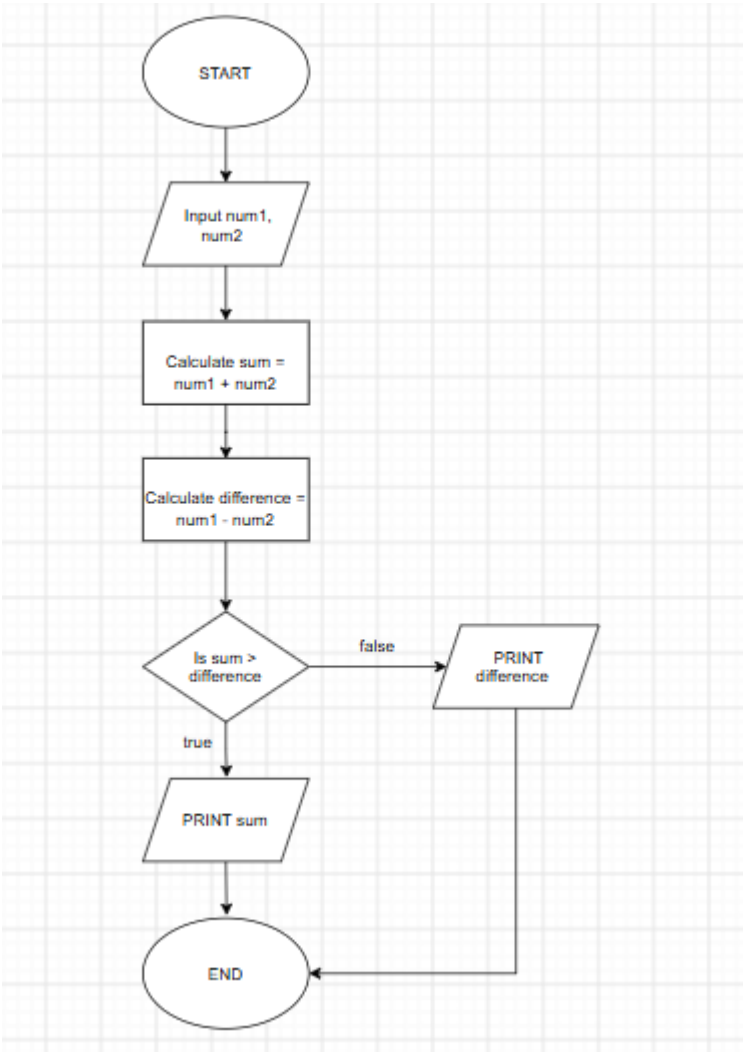
- If No (False), print difference

c) Output : Display Sum

Step 2 : Pseudocode

1. Start
2. Display “Enter two numbers”
3. Input num1 and num2
4. Calculate sum = num1 + num2
5. Calculate difference = num1 – num2
6. If sum > difference, display sum
7. If sum < difference, display different
8. End

Step 3 : Flowchart



4. This is a program to calculate and display the body mass index (BMI). The BMI is to determine whether a person is overweight, underweight or has an optimal weight for his/her height. A person's BMI is calculated using the formula BMI = weight / height2 . A sedentary person's weight is considered to be optimal if his/her BMI is between 18.5 and 25. If the BMI is less than 18.5, the person is considered to be underweight. If the BMI value is greater than 25, the person is considered to be overweight.To determine the BMI, let the organiser enters the total number of people participating in this program. Then let each participant enters his/her weight (in kg) and height (in meters); and as a result, the participant gets a message indicating whether he/she has optimal weight, is underweight, or is overweight. Repeat the process for all participants.

Step 1 : Analysis

a) Input : Total number of participants, and for each participant: weight (in kg) and height (in meters).

b) Process :

i) Loop through the process for each participant based on the total number entered.

ii) Calculate BMI using the formula: BMI $=$ weight$/($height $\times$ height$)$.

iii) Use a decision structure (conditional statements) to evaluate the BMI:

- If BMI < 18.5, classify as Underweight.

- If $18.5 \leq$ BMI ≤ 25, classify as Optimal weight.

- If BMI > 25, classify as Overweight.

c) Output : Display a message indicating whether the participant is underweight, has optimal weight, or is overweight.

Step 2 : Pseudocode

1. Start

2. Display "Enter total number of participants:"

3. Input totalParticipants

4. For i = 1 to totalParticipants Do

5. Display "Enter weight (in kg) and height (in meters):"

6. Input weight, height

7. Calculate bmi = weight / (height * height)

8. If bmi < 18.5 Then

9. Display "Status: Underweight"

10. Else If bmi <= 25 Then

11. Display "Status: Optimal weight"

12. Else

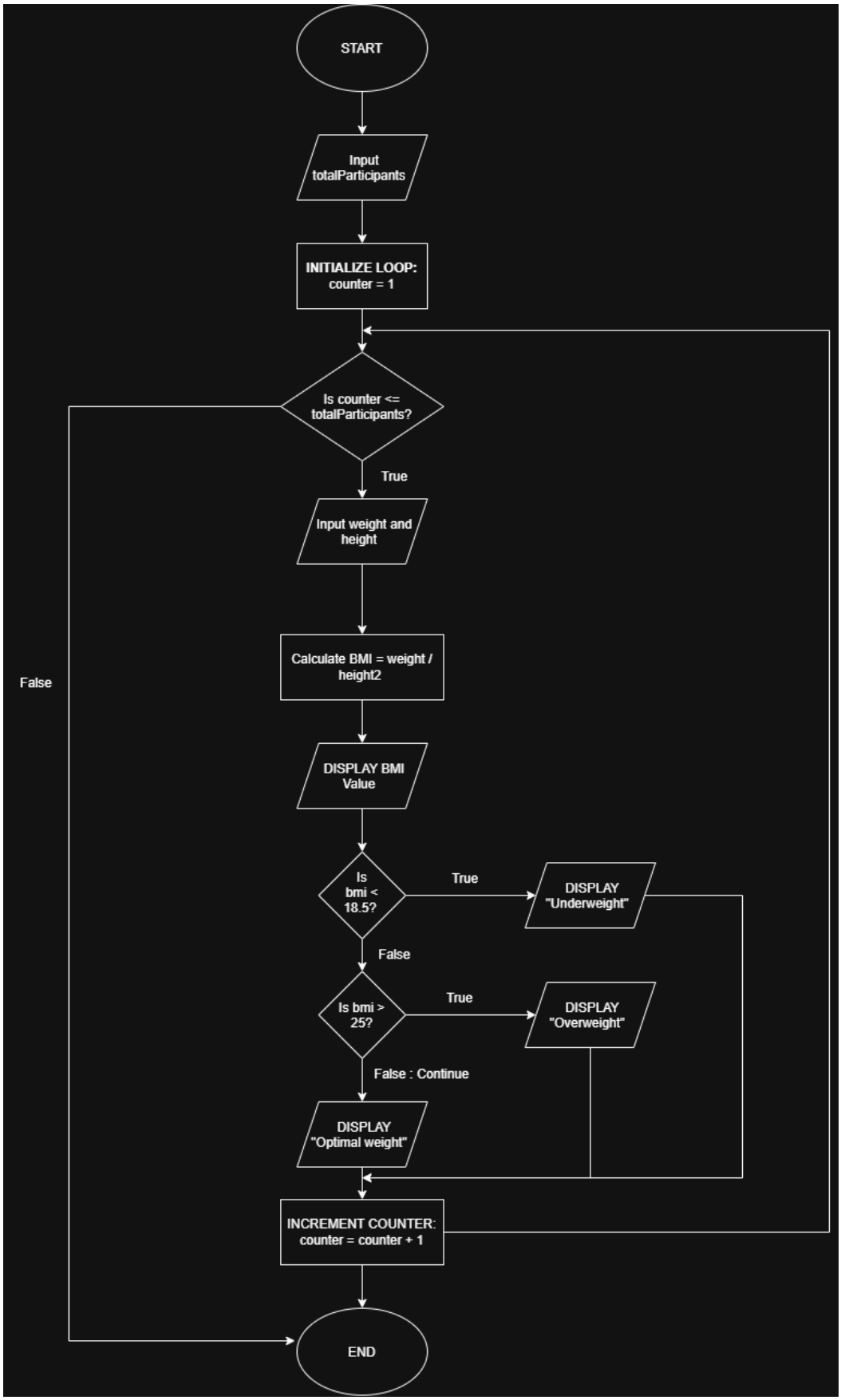
13. Display "Status: Overweight"

14. End If

15. End For

16. End

Step 3 : Flowchart



5. Insert a few integer numbers between -100 and 100 (insert the value 0 to stop). Determine how many negative numbers and how many positive numbers have been inserted, the smallest number and the biggest number, their total and average. Print out what have been determined. (Consider to use the top-down approach in your design).

Step 1 : Analysis

a) Input : Sequence of integers between -100 and 100 (0 stops the input).

b) Process :

i) Accept numbers in a loop until 0 is entered.

ii) Count how many numbers are positive and negative.

iii) Find the smallest and biggest numbers.

iv) Calculate the total sum and average of all numbers.

c) Output : Display counts, smallest, biggest, total sum, and average.

Step 2 : Pseudocode

1. Start

2. Initialize posCount = 0, negCount = 0, sum = 0, count = 0

3. Display "Enter integer (0 to stop):"

4. Input num

5. Set smallest = num

6. Set biggest = num

7. While num != 0 Do

8. Set sum = sum + num

9. Set count = count + 1

10. If num > 0 Then

11. Set posCount = posCount + 1

12. Else If num < 0 Then

13. Set negCount = negCount + 1

14. End If

15. If num < smallest Then Set smallest = num

16. If num > biggest Then Set biggest = num

17. Input num

18. End While

19. If count > 0 Then

20. Set average = sum / count

21. Display posCount, negCount, smallest, biggest, sum, average

22. Else

23. Display "No numbers entered."

24. End If

25. End

Step 3 : Flowchart

